

Attachable Mobile Tail: A Notification System to Convey Emotional Information Intuitively

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Abstract—This study proposes a system in which a tail-shaped device responds to smartphone message notifications by conveying emotions intuitively. Messages are analyzed for sentiment and classified into three polarities: positive, neutral, and negative. Corresponding tail movements are generated based on the results. A user study confirmed that participants could intuitively interpret the emotions from the tail’s movement, demonstrating the effectiveness of this non-verbal notification method.

Index Terms—Human-Machine interaction, tail, emotional expression

I. INTRODUCTION

In contemporary society, smartphones have become deeply integrated into daily life as a primary means of communication. With the widespread use of messaging apps and social networking services, people are constantly surrounded by various notifications. However, these notifications typically rely on inorganic methods such as standard vibrations or alert sounds, making it difficult to intuitively convey the emotional weight or contextual nuances of the information.

In contrast, animal tails serve as natural and expressive interfaces for emotional communication. For example, a dog’s tail clearly reflects emotions such as joy or anxiety through its movement, which humans can intuitively understand [1]. This suggests the potential of tail-like movements as a novel interface for human-device interaction.

In this study, we propose a system that attaches an organic and emotional interface, called a “tail,” to a smartphone. As shown in Figure 1, the tail moves differently in response to message notifications, expressing the emotions contained in the message.

This paper presents the design and implementation of the proposed system, along with the results of a user study evaluating the perceived emotional expressiveness of the tail movements. The findings offer insights into the effectiveness of tactile-anthropomorphic interfaces in enhancing emotional communication through mobile devices.



Fig. 1. Image of System

II. RELATED WORK

As prior research related to this study, we introduce animal tail-inspired interfaces and their applications.

Singh and Young proposed an animal-inspired interface that conveys a robot’s internal state through tail movements similar to those of dogs or cats [2] [3]. They developed a system in which a robot is equipped with a tail to non-verbally communicate its state to the surrounding environment, and reported that users were able to understand emotional nuances through the vibration patterns of the tail.

Attempts have also been made to animate animal-like tails in coordination with smartphones. For instance, the DIGITaIL by The Tail Company in the UK is a tail that responds to smartphone notifications [4]. It vibrates slightly for email alerts and more vigorously for incoming calls, conveying information through tail movements based on the type of notification. These are examples of implementations where internal motors are controlled by apps or notification signals on the smartphone.

Building on this, the present study proposes an intuitive notification system in which animal-like tail movements are presented to the user in a chat application, based on the emotional content of incoming messages.

III. SYSTEM

An overview of the system is shown in Figure 2, and the appearance of the device is shown in Figure 3. A tail device is connected to a smartphone, which can receive audio signals from the smartphone. The tail device includes a built-in servo motor that enables

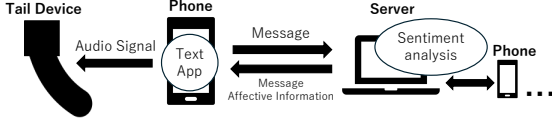


Fig. 2. System Overview

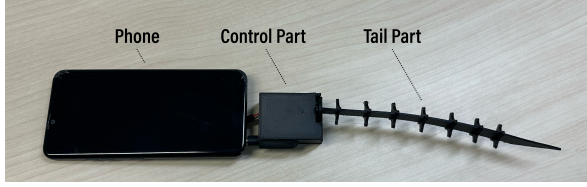


Fig. 3. Device Appearance

it to move the tail accordingly. The smartphone runs a custom chat application. When a message is received, a server performs sentiment polarity classification on the corresponding text data. The sentiment result is returned along with the message. Based on the identified polarity, the smartphone emits an audio signal that is received by the tail device, allowing it to move the tail in response to the message.

In this demonstration, the sentiment polarity of the text data is classified into three categories: negative, neutral, and positive. The tail movement is then controlled according to the classified sentiment.

IV. EXPERIMENT

To confirm that the tail attached to an inanimate object (a smartphone) in this system can express emotions through its movement, the following experiment was conducted.

Participants were shown three types of tail-wagging behaviors and were asked to rate their impressions on a 7-point Likert scale, ranging from 1 ("Does not apply at all") to 7 ("Applies completely"), based on the items shown in Figure n. These items were selected based on Polarity Classification and Ekman's basic emotions. The three tail-wagging conditions were defined as follows:

- **Condition 1:** Small, trembling shakes
- **Condition 2:** Large, slow swings
- **Condition 3:** Large, quick swings

The experiment had 10 participants.

Table I summarizes the experimental outcomes, presenting the mean ($\pm$ SD) ratings from ten participants for each questionnaire item. Positivity served as the primary outcome, whereas the other emotion scores were explored as indicators for future work.

The data were measured under three within-subject conditions. Prior to the main analysis, normality of the positivity scores was assessed using the Shapiro-Wilk test. The results indicated that normality was violated for Condition 1 ($W = 0.752$, $p = .0037$) and Condition 3 ($W = 0.731$, $p = .0021$). Therefore, non-parametric methods were employed for subsequent analyses.

TABLE I
EVALUATION RESULTS (MEAN $\pm$ SD).

Primary Outcome			
	Condition 1	Condition 2	Condition 3
Positivity	2.2 ± 1.1	3.3 ± 2.0	6.5 ± 0.7
Exploratory Emotional Items			
Anger	5.4 ± 0.9	3.0 ± 1.3	2.2 ± 1.6
Disgust	5.6 ± 0.7	4.1 ± 1.5	1.8 ± 1.3
Fear	4.7 ± 1.8	5.0 ± 1.5	1.6 ± 0.9
Happiness	3.1 ± 1.5	3.3 ± 1.8	6.6 ± 0.5
Sadness	3.1 ± 1.6	4.5 ± 2.0	1.4 ± 0.5
Surprise	4.2 ± 2.0	3.1 ± 0.9	3.1 ± 1.4

A Friedman test was conducted to examine overall differences across conditions, revealing a significant effect of condition, $\chi^2(2) = 12.39$, $p = .0020$.

Post hoc pairwise comparisons using the Wilcoxon signed-rank test with Bonferroni correction revealed significant differences between Condition 1 and Condition 3 ($p < .05$) as well as between Condition 2 and Condition 3 ($p < .05$).

In addition, it was observed that the "Anger" item scored higher in condition 1, "Sadness" in condition 2, and "Happiness" in condition 3 compared to other conditions. In this demonstration, emotional expression is based solely on positivity; however, future work may explore the use of a greater variety of movement patterns to enable more complex emotional expressions.

V. DEMONSTRATION

In this demonstration, a smartphone connected to two tail devices is used. Participants operate a chat application on the smartphone to send messages, and in response, the tail devices move to express the emotion conveyed in the message.

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